

AI-Based Text Summarization Platform

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**GitHub Repository:
<https://github.com/himankbhardwaj21/AI-text-summarizer-project>**

Abstract

The AI-Based Text Summarization Platform is a full-stack web application developed to generate concise summaries from large textual content and PDF documents using Artificial Intelligence. The platform allows users to manually enter text or upload PDF files. The backend extracts text from uploaded documents and sends the content to an AI model using API integration. The generated summary is then displayed on the frontend interface. The project demonstrates concepts of frontend and backend development, API integration, file handling, and Natural Language Processing (NLP).

Introduction

Large amounts of textual information are generated daily through articles, blogs, research papers, and online documents. Reading lengthy content can consume significant time and effort. Automatic text summarization helps reduce large text into concise and meaningful summaries. The AI-Based Text Summarization Platform was developed to solve this problem using Artificial Intelligence and Natural Language Processing techniques. The application provides a simple user interface where users can either enter text manually or upload PDF files for summarization. The project also demonstrates practical implementation of API integration, PDF text extraction, and full-stack application development.

Objectives

- Develop an AI-powered text summarization system.
- Accept manual text input from users.
- Support PDF document upload and text extraction.
- Generate concise summaries using AI APIs.
- Build a responsive and user-friendly interface.
- Implement backend API communication and file handling.

Technology Stack

Component	Technology Used
Frontend	React.js
Styling	Tailwind CSS
Backend	Node.js + Express.js
AI Integration	OpenRouter API
PDF Extraction	pdf-parse
File Upload	multer
HTTP Requests	axios

System Architecture

The platform follows the workflow shown below: User Input → Frontend → Backend API → AI Model → Summary Output

1. User enters text or uploads a PDF file.
2. Frontend sends request to backend server.
3. Backend extracts text from uploaded PDF.
4. Backend sends text to AI model using API integration.
5. AI model generates summarized content.
6. Summary is displayed on the frontend interface.

Features

- AI-generated text summarization
- PDF upload support
- PDF text extraction
- Responsive frontend interface
- Copy summary functionality
- Clear text functionality
- Error handling and validation

Implementation

The frontend was developed using React.js and Tailwind CSS to create a clean and responsive user interface. The backend server was developed using Node.js and Express.js. The application uses multer for file upload handling and pdf-parse for extracting text from uploaded PDF documents. OpenRouter API was integrated to generate summaries using AI models. Axios was used to handle communication between frontend and backend services. Error handling was implemented to manage invalid input, file issues, and API errors.

Results

The developed application successfully generates summaries from both manually entered text and uploaded PDF documents. The project demonstrates successful integration of AI APIs with a full-stack web application. The platform provides a user-friendly interface and performs summarization efficiently for educational and practical purposes.

Future Scope

- Multi-language summarization
- Download summary as PDF
- Chat with PDF documents
- User authentication system
- Summary length customization
- Voice-based summarization

Conclusion

The AI-Based Text Summarization Platform was successfully developed using modern web technologies and AI integration techniques. The project helped demonstrate concepts of frontend-backend communication, Natural Language Processing, API integration, PDF handling, and full-stack development. The platform is capable of generating concise summaries from large text and PDF documents efficiently.

References

- React.js Documentation
- Node.js Documentation
- Express.js Documentation
- Tailwind CSS Documentation
- OpenRouter API Documentation
- GitHub Repository: <https://github.com/himankbhardwaj21/AI-text-summarizer-project>